Java i

Ushtrime Numerike

Sisteme dhe Sinjale

* Sinjali:

Në inxhinieri elektrike, sasia themelore e paraqitjes së informatës quhet sinjal, pasi parashtë a është informacion digjital apo analog. Matematikisht sinjali paraqet funksion që bartë informacion.

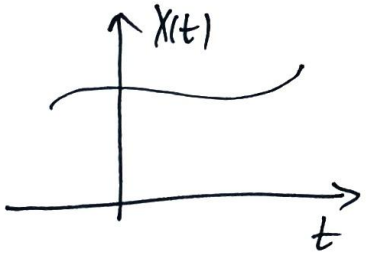
* Sistemi:

Sistem quajm një medium fizik të cilët në hyrjen x(t), x(n), përpunojnë me dalje y(t), y(n).

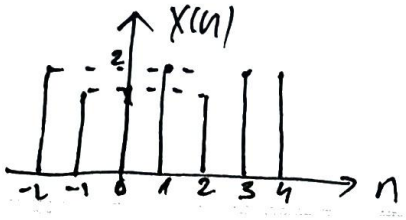
* Sinjalet klasifikohen në dy grupe të mëdha:

1. Sinjalet kontinuale
2. Sinjalet diskrete

* Sinjali kontinual është sinjali i përkufizuar për çfarëdo vlere të "t"; pra $t \in R$.

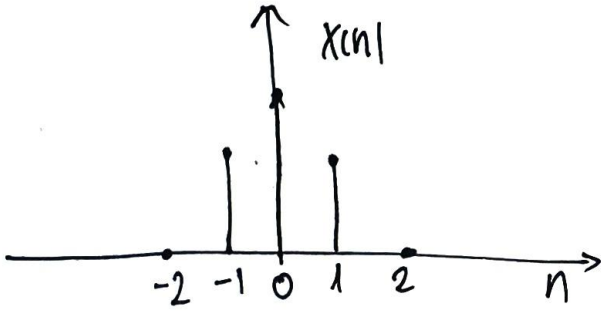


* Sinjali diskret është sinjali i përkufizuar vetëm për vlera të plota të variablit "n"; pra $n \in Z$

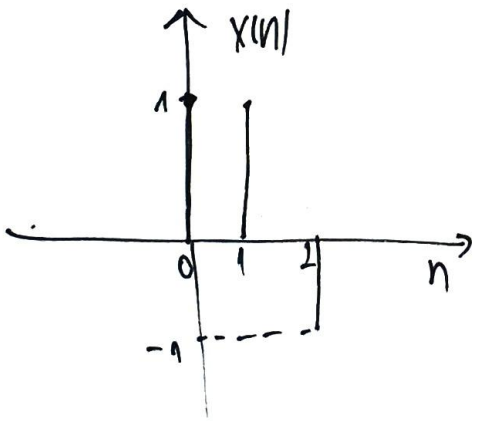


Sinjalet diskrete mund të jepen në formë si:

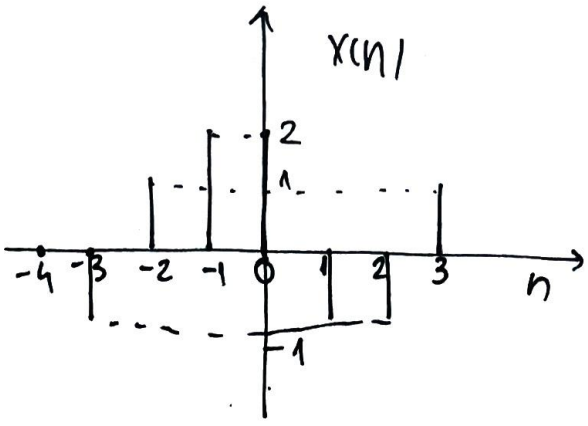
1° $x(n) = \{0, 1, \underset{\uparrow}{2}, 1, 0\}$ → shigjeta tregon fillimin e mostrës nga zero.



2° $x(n) = \{1, 1, -1\}$



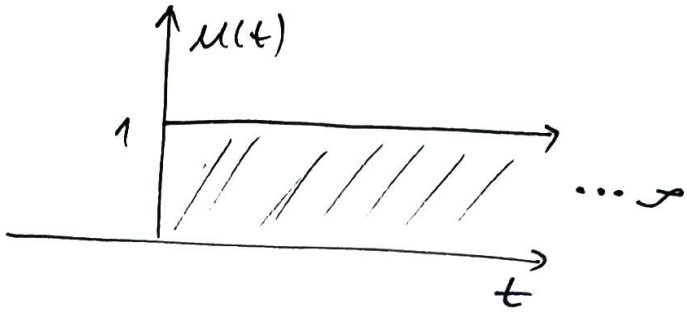
Sh.



⇒ $x(n) = \{0, -1, 1, 2, \underset{\uparrow}{2}, -1, -1, 1\}$

Sinjali "shkallë" - njësi

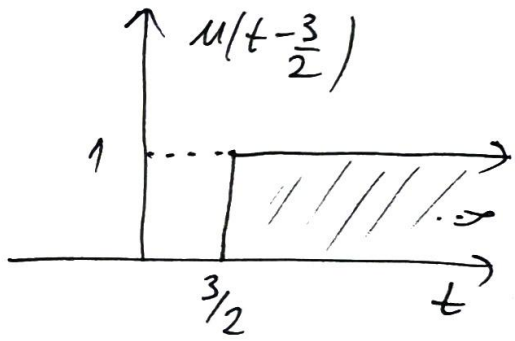
$$u(t) = \begin{cases} 1 & t \geq 0 \\ 0 & t < 0 \end{cases}$$



Shembuj

1° $u\left(t-\frac{3}{2}\right) = \begin{cases} 1 \\ 0 \end{cases}$ $\quad t-\frac{3}{2} \geq 0 \Rightarrow t \geq \frac{3}{2}$

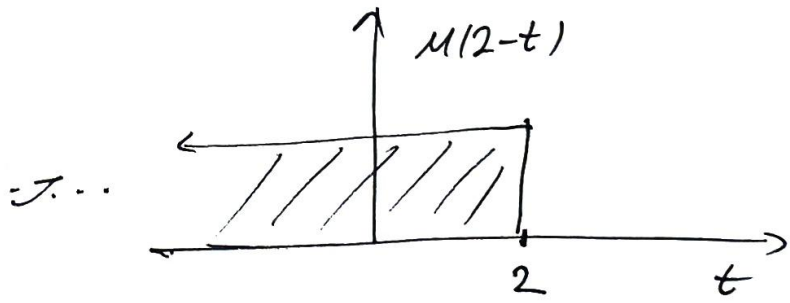
$t < \frac{3}{2}$



2° $u(2-t) = \begin{cases} 1 \\ 0 \end{cases}$ $\quad 2-t \geq 0 \Rightarrow -t \geq -2$

$\boxed{t \leq 2}$

$t > 2$



Shembuj:

Skiconi sinjalet:

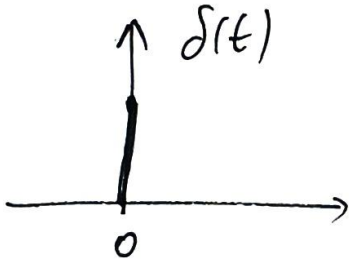
a) $u(t-3)$

b) $u\left(\frac{1}{2}-t\right)$

c) $u(t+4)$

Sinjali i Diraluet (delta impulsi)

" $\delta(t)$" → simboli



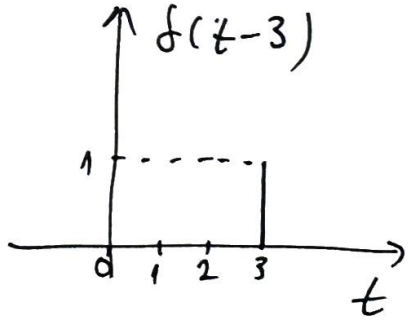
Sinjal me koheZgjatje shume" te" shkurtër dhe amplitud shume" te" madhe

$$\delta(t) = \begin{cases} 1 & t \neq 0 \\ 0 & t = 0 \end{cases}$$

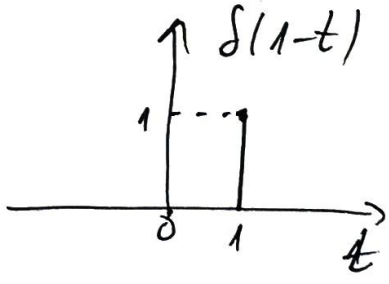
$$\int_{-\infty}^{\infty} x(t)\,\delta(t)\,dt = x(0)$$

shembuj)

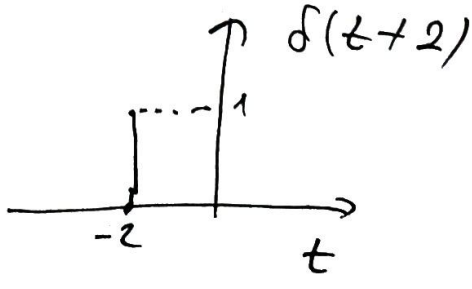
$$\delta(t-3) = \begin{cases} 1 & t-3=0 \Rightarrow t=3 \\ 0 & t \neq 3 \end{cases}$$



$$\delta(1-t) = \begin{cases} 1 & 1-t=0 \\ & t=1 \\ 0 & t \neq 1 \end{cases}$$



$$\delta(t+2) = \begin{cases} 1 & t+2=0 \\ & t=-2 \\ 0 & t \neq -2 \end{cases}$$



Vetit e sinjalit të Diraku:

1° $\delta(-t) = \delta(t)$

2° $\delta'(t) = -\delta'(t)$

3° $\delta(at) = \frac{1}{|a|}\delta(t)$

4° $x(t)\cdot\delta(t) = x(0)\delta(t)$, $t=0$

5° $x(t)\delta(t-t_0) = x(t_0)\delta(t-t_0)$, $t=t_0$

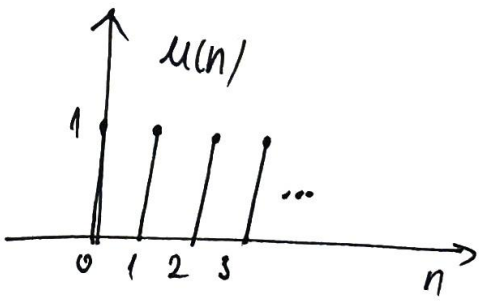
6° $\int_{-\infty}^{\infty} x(t)\delta(t)\,dt = x(0)$

7° $\int_{-\infty}^{\infty} x(t)\,\delta(t-t_0)\,dt = x(t_0)$

Sinjalet Diskrete

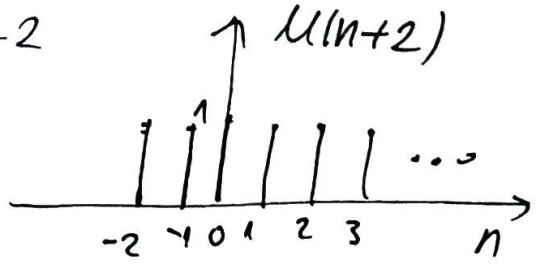
Sinjali shkallë-njësi

$$u(n) = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases}$$

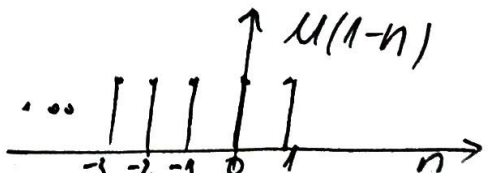


$u(n-n_0)$; $n, n_0 \in \mathbb{Z}$

$$u(n+2) = \begin{cases} 1 & n+2 \geq 0 \Rightarrow n \geq -2 \\ 0 & n < -2 \end{cases}$$

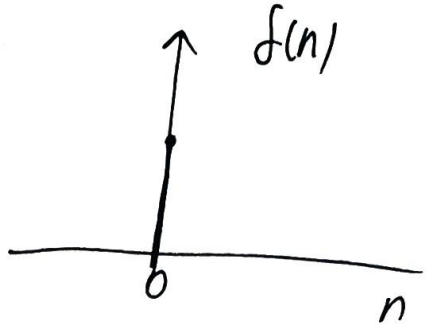


$$u(1-n) = \begin{cases} 1 & 1-n \geq 0 \Rightarrow n \leq 1 \\ 0 & n > 1 \end{cases}$$

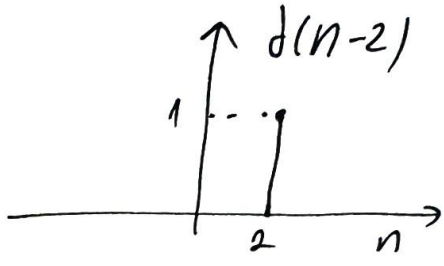


Sinjali i Diracit për diskrete

$$\delta(n) = \begin{cases} 1 & n=0 \\ 0 & n\neq 0 \end{cases}$$



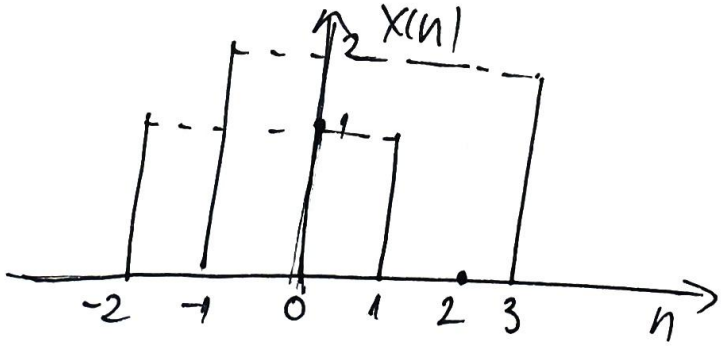
$$\delta(n-2) = \begin{cases} 1 & n-2=0 \\ & n=2 \\ 0 & n\neq 2 \end{cases}$$



$$\boxed{\delta(n) = u(n) - u(n-1)}$$

$$\boxed{\sum_{n=-\infty}^{\infty} x(n)\cdot\delta(n-n_0) = x(n_0)}$$

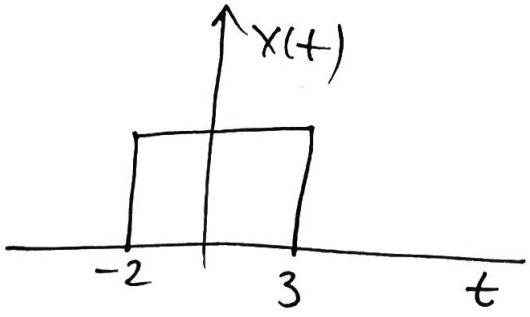
Sh) Nëse "shtë" dhënë figura të" paraqitet shprehja përmes sinjaleve të Diracit



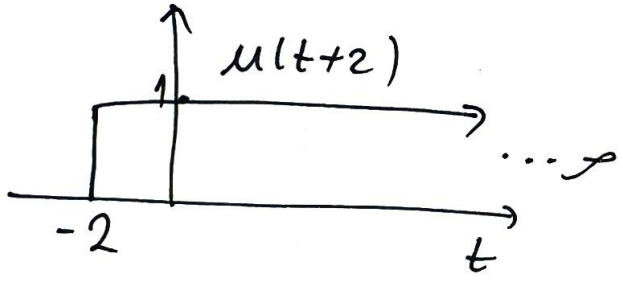
Zgjidhja

$$x(n) = \underset{n=-2}{\delta(n+2)} + 2\underset{n=-1}{\delta(n+1)} - \underset{n=0}{\delta(n)} + \underset{n=1}{\delta(n-1)} + 2\underset{n=3}{\delta(n-3)}$$

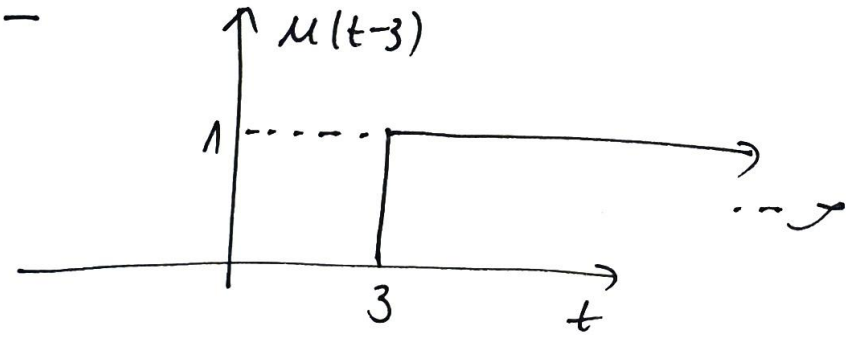
Formimi i sinjalave drejtkëndshe përmes sinjalit "shkallë"- njësi.

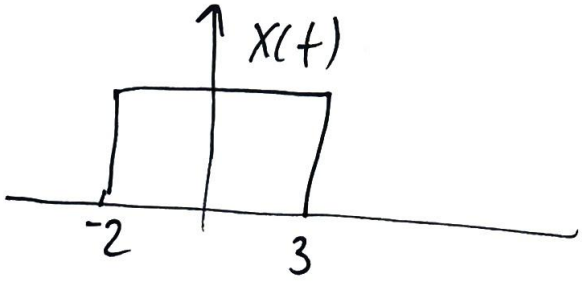


$\Rightarrow x(t) = u(t+2) - u(t-3)$

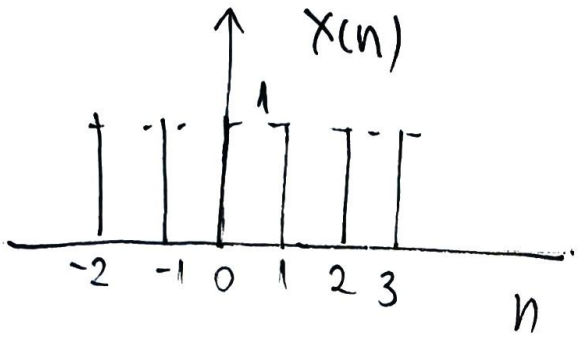


Zbresim

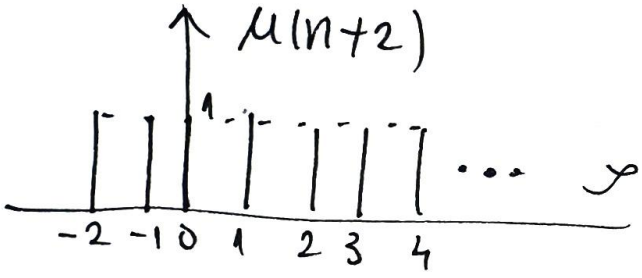


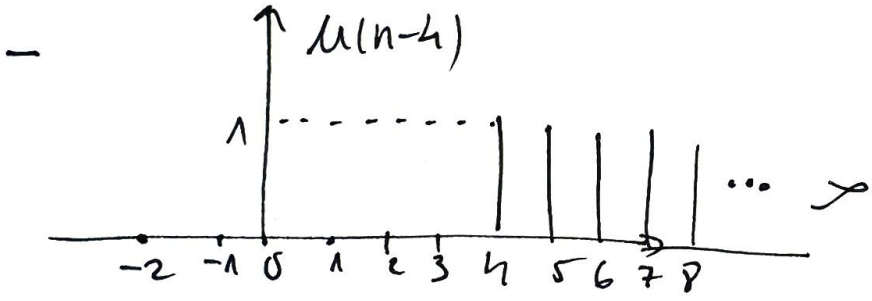


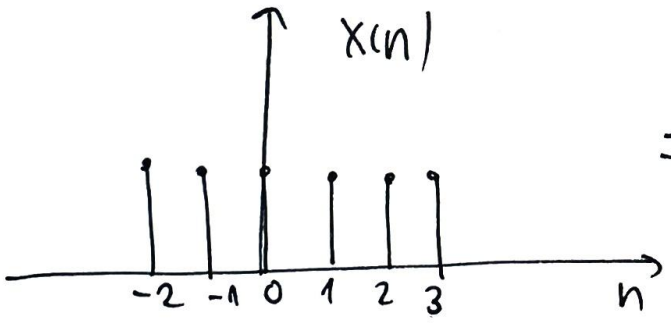
Sinyali Diskret











$\Rightarrow \quad x(n) = u(n+2) - u(n-4)$

1. Paraqitni grafikisht sinjalin:

$$x(t) = \underbrace{(t-5)}_{\text{f-oni}} \underbrace{[u(t) - u(t-4)]}_{\text{shtrirja} \Rightarrow \{0,4\}}$$

Zgjidhja

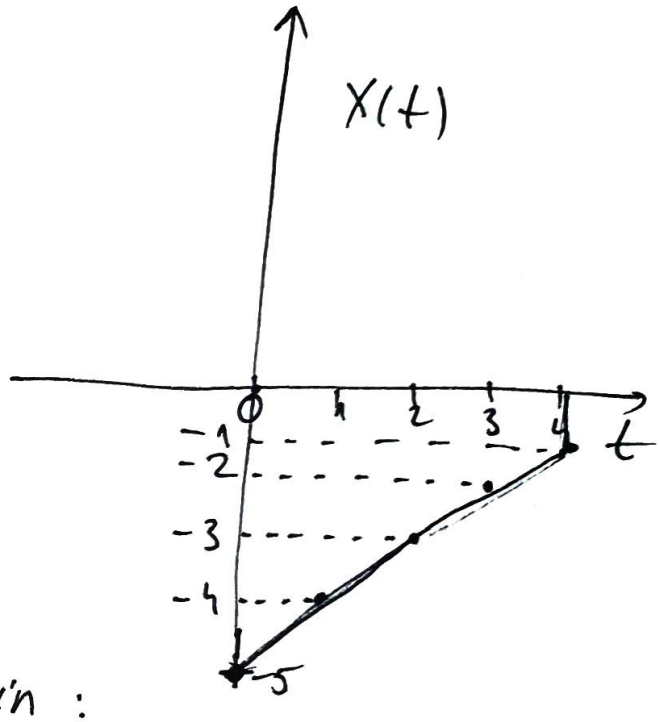
$t=0 \Rightarrow 0-5=-5$

$t=1 \Rightarrow 1-5=-4$

$t=2 \Rightarrow 2-5=-3$

$t=3 \Rightarrow 3-5=-2$

$t=4 \Rightarrow 4-5=-1$



2. Paraqitni grafikisht sinjalin:

$$x(n) = \underbrace{(2^n+1)}_{\text{f-oni}} \underbrace{\{u(n+1) - u(n-4)\}}_{\text{shtrirja} \rightarrow \{-1,3\}}$$

Zgjidhja

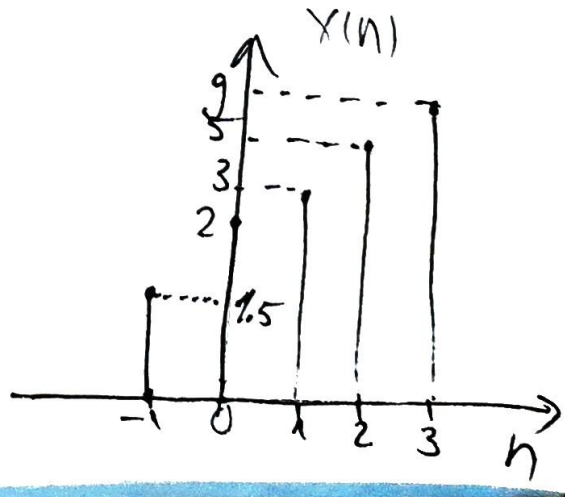
$n=-1 \rightarrow 2^{-1}+1 = \frac{1}{2}+1 = \frac{3}{2}$

$n=0 \rightarrow 2^0+1 = 1+1 = 2$

$n=1 \rightarrow 2^1+1 = 2+1 = 3$

$n=2 \rightarrow 2^2+1 = 4+1 = 5$

$n=3 \rightarrow 2^3+1 = 8+1 = 9$



Vetit e sinjalere kontinuale

1° Zhvendosja në" kohë"

$y_1(t) = Ax(t-t_0)$

a) $x(t+t_0) \rightarrow$ zhvendoset në" të" majtë"

b) $x(t-t_0) \rightarrow$ —11— —11— djathtë

2° Pasqyrimi në" kohë"

$x(-t) \rightarrow$ sinjali rrotullohet rreth boshtit të" amplitudës

3° Shkallëzimi në" kohë" (ngjeshtimi dhe zgjerimi)

a) $x(2t) \rightarrow$ Brezi i sinjalit ngjeshtohet për 2x

b) $x(t/2) \rightarrow$ —11— —11— zgjerohet —11—

4° Shkallëzimi në" Amplitudë"

$y_1(t) = Ax(t)$

Sinjalet Diskrete

3° $x(2n) \rightarrow$ ngjeshtohen vetem pjesët e plota